

ERIC GICHURI MUCHIRI

Backend Developer | Python | Flask | PHP | MySQL | REST APIs

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Portfolio: <https://ericgichuri.github.io/>

GitHub: <https://github.com/ericgichuri>

PROFESSIONAL SUMMARY

Computer Science graduate and hands-on software developer with practical experience building and supporting web-based business applications using **Python, Flask, PHP, JavaScript, MySQL and REST APIs**. Experienced in developing database-driven applications, designing relational data structures, implementing backend business logic, integrating frontend interfaces with APIs, troubleshooting application issues, and delivering software for real-world business requirements.

Particularly experienced in **real-estate and marketplace applications**, including property listings, rental workflows, user accounts and location-based data. Recently completed a client real-estate project using **Python Flask, MySQL, Tailwind CSS and JavaScript/jQuery**.

Strong interest in backend architecture, database engineering, API development and building reliable production-oriented software. Comfortable working independently, understanding business requirements and turning them into functional software solutions.

CORE TECHNICAL SKILLS

- **Backend Development:** Python, Flask, PHP
- **API Development:** REST APIs, Flask APIs, JSON, API integration
- **Databases:** MySQL, SQL, MongoDB
- **Database Design:** Relational modelling, relationships, queries, CRUD operations
- **Frontend Integration:** JavaScript, jQuery, React, HTML, CSS, Tailwind CSS, Bootstrap
- **Authentication:** User authentication and access control implementation
- **Version Control:** Git, GitHub
- **Data & Reporting:** Microsoft Excel, SPSS, STATA
- **Software Development:** Business logic, debugging, system integration, troubleshooting
- **IT:** System support, software installation, hardware troubleshooting, networking fundamentals

- **Development Approach:** AI-assisted development, rapid iteration, debugging and refactoring
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PROFESSIONAL EXPERIENCE

Software Developer / Freelance Project Work

Real-Estate & Web Application Development

2023 – Present

- Design and develop database-driven web applications using **Python, Flask, MySQL, JavaScript and Tailwind CSS**.
- Develop backend routes, APIs, authentication, business logic and database interactions.
- Design relational database structures for applications involving users, properties, listings and related business entities.
- Integrate backend APIs with frontend interfaces using JavaScript/jQuery and JSON.
- Implement property listing and rental-related workflows for real-estate platforms.
- Debug frontend and backend integration issues and improve application reliability and usability.
- Develop reusable backend functionality and organize Flask applications using structured routes and application components.
- Work directly from business requirements to develop, test and refine application features.
- Use Git/GitHub for source-code management and project development.
- Use AI coding tools as development assistants for debugging, code generation, refactoring, SQL work and problem solving while making independent technical decisions.

Selected Project: MRentKenya / Real-Estate Platform

- Developed and enhanced a real-estate platform focused on rental properties and property sales.
- Implemented property listing functionality, user authentication and database-driven property information.
- Worked with property-related entities and workflows including properties, listings, landlords and rental information.
- Developed backend functionality using **Python Flask and MySQL**, with JavaScript/jQuery and Tailwind CSS on the frontend.
- Built API endpoints to provide structured property information to frontend components.
- Implemented search, filtering and property-detail functionality.
- Integrated frontend interfaces with backend APIs and resolved application and UI integration issues.
- Recently completed a client project version of the platform using **Flask, Tailwind CSS and JavaScript/jQuery**.

Computer Packages Instructor / IT Support

Okoka Humanitarian CBO, Ruiru

February 2024 – December 2024

- Trained users in Microsoft Excel, Word, Access and PowerPoint.
- Provided technical support and troubleshooting for organizational computer systems.
- Assisted users in resolving software and system-related issues.
- Supported website updates and maintenance.
- Prepared and maintained Excel-based records including attendance, fees, balances and timetables.
- Assisted users with digital tools and provided practical guidance on software usage.
- Supported day-to-day technology operations and helped minimize interruptions to organizational activities.

Collection Agent

Bunnitech Collection Ltd, Ruiru Bypass

June 2023 – December 2023

- Maintained accurate customer and loan repayment records.
- Performed data entry and ensured customer information remained up to date.
- Analysed client information to prioritize collection activities.
- Prepared and maintained accurate records for reporting.
- Communicated with customers through calls and messages while maintaining professional records.

Attachment — Customer Support / Operations

Amica Savings & Credit, Murang'a

April 2022 – July 2022

- Supported customers with mobile banking services and related processes.
- Assisted with data entry and document management.
- Maintained organized customer records and operational documentation.
- Supported branch operations and assisted customers with service-related requests.
- Worked with staff to improve the handling and submission of customer documentation.

SELECTED SOFTWARE PROJECTS

MRentKenya — Real-Estate / Rental Platform

Python, Flask, PHP, MySQL, JavaScript, jQuery, Tailwind CSS

- Real-estate platform supporting property listings and rental-related workflows.
- Developed backend functionality, database integration and API-driven frontend features.
- Worked with property and user-related data and application business logic.
- Implemented authentication and property listing functionality.
- Developed and maintained frontend/backend integration.

Website: <https://mrentkenya.co.ke> soon to migrate to this python flask <https://beta.mrentkenya.co.ke/>

Kenya Geo API — REST API

Python / API Development

- Developed a RESTful API providing structured geographical data for Kenya.
- Designed API endpoints for counties, constituencies and wards.
- Structured data for consumption by web applications and other clients.
- Practiced API development, JSON responses and backend data organization.

Website: <https://kenya-geo-api.vercel.app>

ESSPOS — Point of Sale & Inventory System

Python, Flask, MySQL, JavaScript, Tailwind CSS

- Developed a database-driven POS and inventory management application.
- Designed backend functionality around products, inventory, users, sales and operational workflows.
- Implemented role-based application functionality for different system users.
- Developed API endpoints and integrated them with frontend components.
- Worked with MySQL database structures and SQL queries.

- Implemented operational features including till sessions, inventory management and transaction workflows.
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ESS EduSystem — School Management System

Python, Flask, CustomTkinter, SQLAlchemy, MySQL

- Developed a school management application with a Flask API backend.
 - Worked with relational data involving students, guardians, staff, attendance, examinations and fees.
 - Used SQLAlchemy and MySQL for database operations.
 - Implemented application modules, user roles and system workflows.
 - Developed desktop and web components for managing school information.
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EDUCATION

Bachelor of Science in Computer Science

Murang'a University of Technology

2019 – 2023

Second Class Upper Division

Kenya Certificate of Secondary Education

Tigithi Boys High School

2015 - 2018

Grade: C+

CERTIFICATIONS

- **Cisco Networking Essentials** — Cisco, 2022
 - **Introduction to Data Science** — Cisco, 2023
 - **Computer Hardware Basics** — Cisco, 2023
 - **SPSS Statistical Package** — KESAP Research Center, 2020
 - **STATA Statistical Software** — KESAP Research Center, 2020
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ADDITIONAL STRENGTHS

- Strong practical experience building software from business requirements.

- Experience working independently on complete application features.
 - Experience developing software for real-world business use cases.
 - Practical understanding of relational databases and SQL.
 - Comfortable debugging backend and frontend integration issues.
 - Experience supporting and training end users.
 - Strong interest in backend architecture and scalable application development.
 - Able to learn new frameworks, technologies and business domains quickly.
 - Comfortable working with AI-assisted development tools while retaining responsibility for technical decisions and code quality.
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REFERENCES

George Kabuu

Branch Manager, Amica Savings & Credit
0727801359

Harun Waithaka

Office Administrator, DCC Laikipia Central Sub County
0728093716